

Content Provenance and Authenticity

Embed trust into media with verifiable provenance and authenticity

Challenges of truth in the genAI era

Growing danger of disinformation

AI generated media is already prevalent

Loss of source info during sharing

Original “who made this” information often disappears making it easy for false or manipulated versions to spread

Accessibility to verification technology

Consumers must easily grasp the tools to fight disinformation

Erosion of trust in information

Bad actors using branded content to make disinformation seem legitimate

What's content provenance?

Verifying origin, tracking modifications, and ensuring integrity throughout digital media's lifecycle

What's C2PA?



Coalition for
Content Provenance
and Authenticity

- Open standard for proving digital content origin and authenticity
- Enables chain of trust from content creation source to consumption using cryptographic signatures and verified metadata
- Helps governments, journalists, and consumers track history and authenticity of videos/images

C2PA capabilities & limits (early 2026)

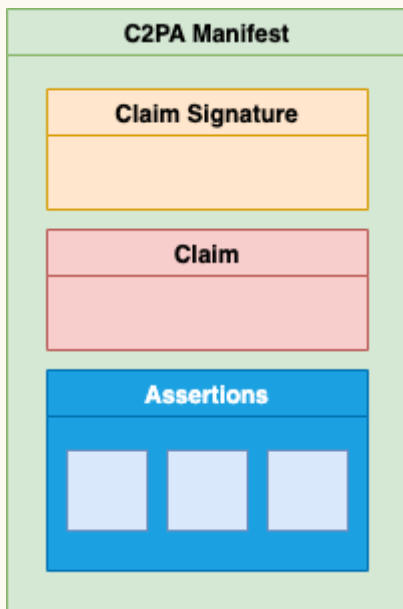
- Defines elaborate signed metadata format (C2PA manifest) to capture provenance record from creation of media through edits to publication
- Defines 'hard binding' capabilities to link metadata to (some) media formats – typically embedded in the file header
- Enables 'soft binding' to link metadata to forensic watermarks or fingerprints
- Embedded metadata can grow very large over time

What's in a C2PA manifest?



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A structured, signed record of a media asset's origin, history, and trust signals



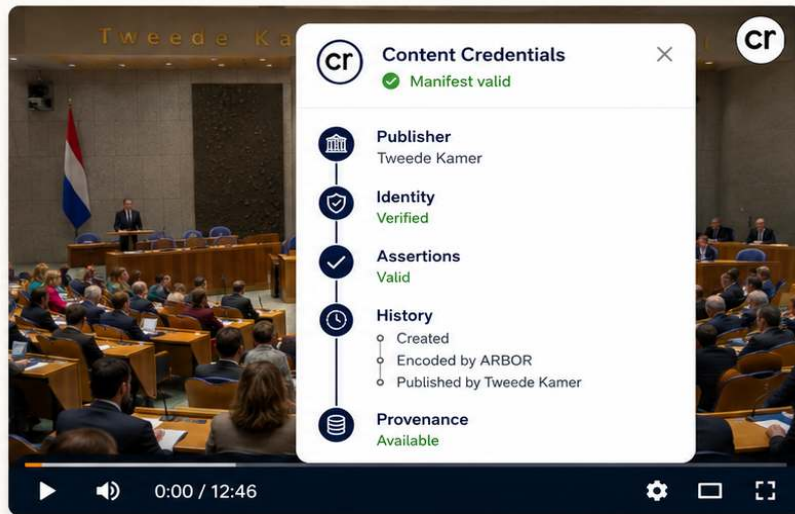
- A C2PA manifest is a signed metadata record attached to a photo or video
- It can describe who created, published, or modified the content, and when
- It can carry information such as publisher identity, timestamps, tools used, and related assertions
- It gives validators something structured to check when assessing authenticity and provenance

Validation: what the verifier sees

Validation should turn technical provenance signals into a clear and understandable trust outcome



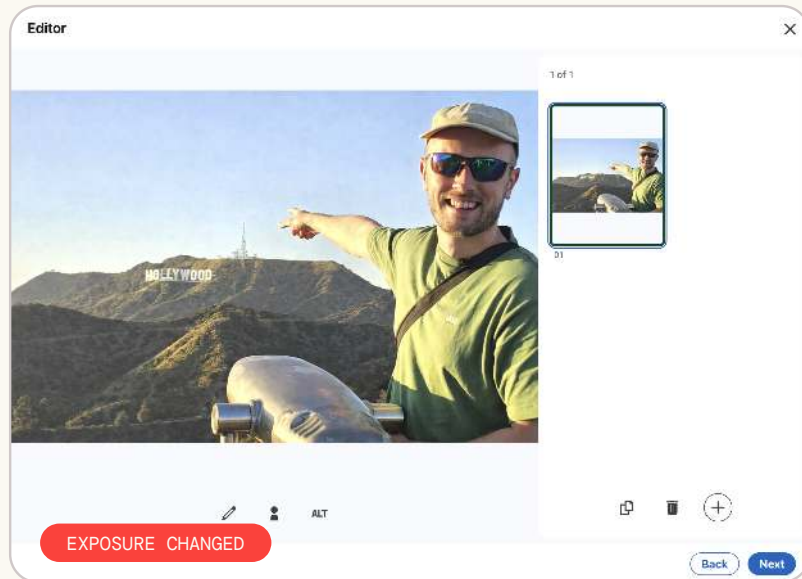
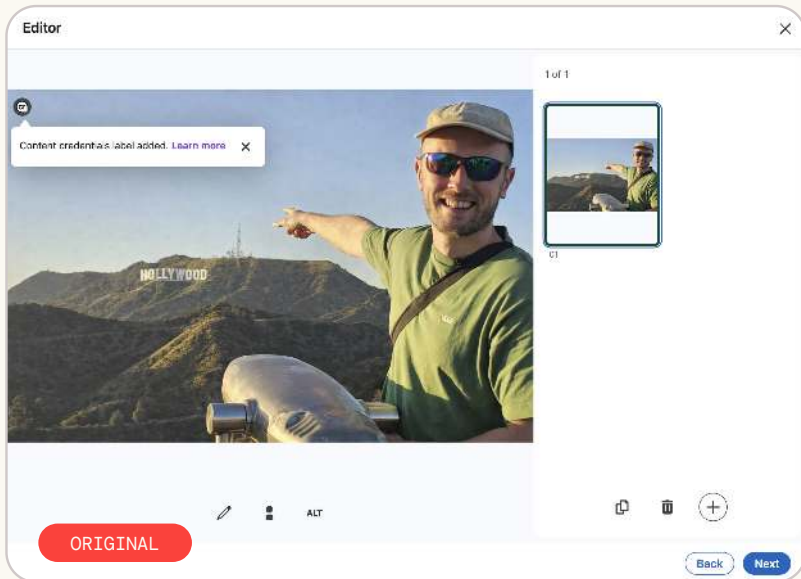
Viewers can see the Content Credentials symbol
(interactive indicator)



Viewer can inspect manifest validation details
(publisher, identity, assertions, history, provenance)

C2PA limits (early 2026)

Example challenge: Easily broken as changing image exposure can strip metadata (LinkedIn)



Provenance implementation methods

Metadata in file

All intermediaries handling media must completely support the embedded metadata carriage to ensure it isn't lost.

Invalidates with any media transform (aka 'hard binding') and easy to strip.

Example technology touchpoints:

- Codecs (bitstreams) or containers
- Media processing (editors, transcoders, etc.)
- Transmission (players, browsers, social media)

Metadata linked via watermark

Provenance information is mostly agnostic from the media technology used for processing and transport.

More robust against media transforms (aka 'soft binding') and difficult to strip, but still requires workflow integration for capturing edits.

Example:

- Watermark survives edit, but edit doesn't update linked metadata

Forensic watermarking approach

A layer of imperceptible data hidden in media that can be verified by simply taking a photo or screenshot of the content (similar to scanning a QR code)



CONTAINS HIDDEN ID DATA LINKING TO
AUTHENTICITY CREDENTIALS

Deepfake detection example



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